



SYDNEY TECHNICAL HIGH SCHOOL

HSC Assessment Task 2

March 2009

Name: _____ Teacher: _____

Mathematics

Time allowed —70 minutes

Instructions

- Approved calculators may be used.
- All necessary working must be shown. Marks may not be awarded for careless or badly arranged work.
- Marks awarded are shown on each question.
- Total marks — 50
- Attempt all questions.
- Start each question on a new page.

Question	1	2	3	4	5	Total /50	%
Marks /10							

Question 1

Marks 10

- a) The curve $y = ax^2 + 4x - 5$ has a gradient of 10 when $x = 2$. 2
Find the value of a .
- b) Find the domain over which the curve $y = x^3 - x^2 - 8x + 4$ is increasing. 2
- c) Use calculus to identify and determine the nature of any stationary points 6
and points of inflection of the function $y = x^3 - 12x$.
Hence, sketch the curve.

Question 2

Marks 10

- a) (i) Determine whether $y = x^7$ is an odd or even function or neither. 1
- (ii) Hence evaluate $\int_{-3}^3 x^7 dx$. 1
- b) The efficiency, E percent, of a particular spark plug when the gap is set to 4
 x mm, is given by $E = 800x - 1600x^2$.
Find the gap setting which gives maximum efficiency.
- c) Two circles have radii a and b cm such that $a + b = 16$. 4
Find the minimum sum of their areas.

Question 3

Marks 10

a) Find the following integrals:

(i) $\int (2x + 3)^2 dx$ 2

(ii) $\int \frac{1}{x^2} dx$ 2

b) (i) Consider a quadrant of a circle of radius 2 units. Dividing the area of the quadrant into five equal sub-intervals produced the following table of ordinates. 4

x	0	0.4	0.8	1.2	1.6	2.0
y	2	1.96	1.83	1.60	1.20	0

Use the Trapezoidal Rule to find an approximate area of the quadrant.
(Write your answer to two decimal places.)

(ii) Calculate the area of a quadrant of a circle of radius 2 units using the formula $A = \pi r^2$. 1

(iii) Explain why your approximation of the area is an underestimate. 1

Question 4

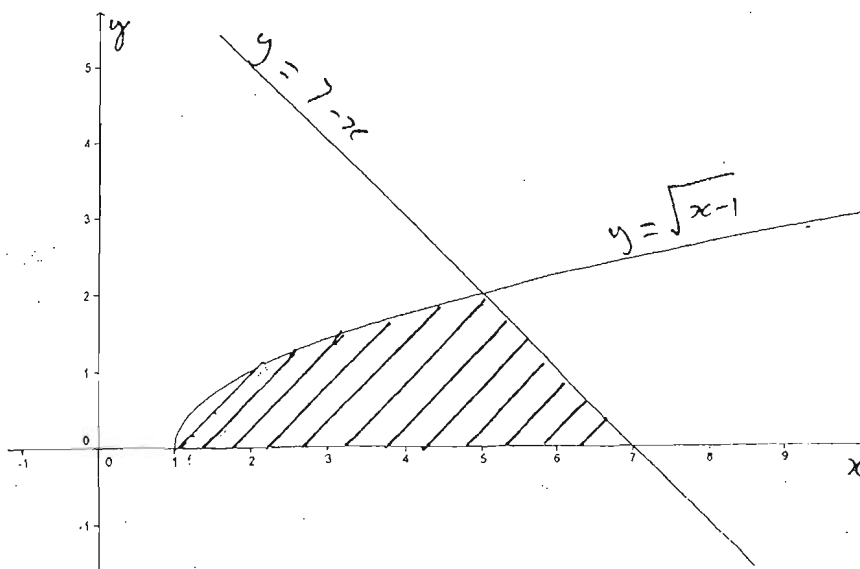
Marks 10

- a) Find the area between the curve $y = x^2 + 4$, the x-axis and the ordinates $x = 2$ and $x = 4$. 2
- b) Calculate the area between the curve $y = x(x+1)(x-2)$, the x-axis and the ordinates $x = 1$ and $x = 3$. 4
- c) What is the area between the curve $y = \sqrt{1-x}$ and the y-axis, between the ordinates where $x = 0$ and $x = \frac{3}{4}$? 4

Question 5

Marks 10

- a) Find the area of the region bounded by the graphs of $y = x^2$ and $y = x^3$. 3
- b) A parabolic mirror is made by revolving the area bounded by the parabola $y = \frac{1}{2}x^2$, the y-axis and the line $y = 4$, about the y-axis. 3
What volume does it occupy?
- c) Calculate the volume when the shaded area is revolved around the x-axis. 4



MATHEMATICS ASSESSMENT TASK 2 SOLUTIONS 2009

Question 1

$$a) y = ax^2 + 4x - 5$$

$$\therefore \frac{dy}{dx} = 2ax + 4 \rightarrow ①$$

$$\text{gradient} = 10 \text{ when } x = 2$$

$$\therefore 2a \cdot 2 + 4 = 10$$

$$\therefore 4a = 6$$

$$\therefore a = \frac{3}{2} \rightarrow ①$$

$$b) y = x^3 - x^2 - 8x + 4$$

$$\therefore \frac{dy}{dx} = 3x^2 - 2x - 8$$

$$= (3x + 4)(x - 2)$$

$$= 0 \text{ when } x = -\frac{4}{3}, 2 \rightarrow ①$$

$$\text{leading coeff of } \frac{dy}{dx} > 0$$

$$\therefore \frac{dy}{dx} > 0 \text{ when } x < -\frac{4}{3} \text{ or } x > 2 \rightarrow ①$$

$$c) y = x^3 - 12x$$

$$\therefore y' = 3x^2 - 12 \rightarrow ①$$

$$= 3(x^2 - 4)$$

$$= 3(x - 2)(x + 2)$$

$$= 0 \text{ when } x = +2, -2 \rightarrow ①$$

$$y = -16, 16 \rightarrow ①$$

$$y'' = 6x$$

$$> 0 \text{ when } x = 2$$

$$\therefore \text{min. at } (2, -16) \rightarrow ①$$

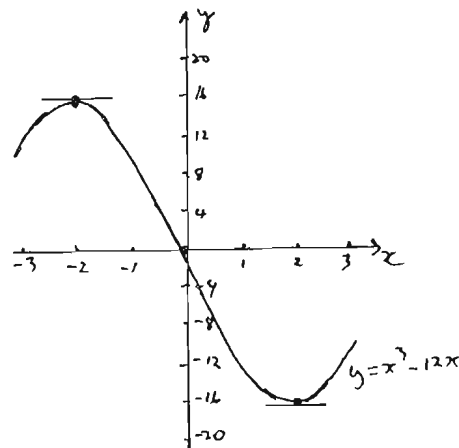
$$< 0 \text{ when } x = -2$$

$$\therefore \text{max. at } (-2, 16) \rightarrow ①$$

$$= 0 \text{ when } x = 0$$

$$y' \neq 0 \text{ when } x = 0$$

$$\therefore \text{a point of inflection at } (0, 0) \rightarrow ①$$



Question 2

$$a) (i) \text{ let } f(x) = x^7$$

$$f(-x) = (-x)^7$$

$$= -x^7$$

$$= -f(x) \rightarrow ①$$

$$\therefore f(x) \text{ is an odd function.}$$

$$(ii) \int_{-3}^3 x^7 dx = 0 \rightarrow ①$$

$$\text{as } f(x) \text{ is odd.}$$

$$b) E = 800x - 1600x^2$$

$$\therefore \frac{dE}{dx} = 800 - 3200x \rightarrow ①$$

$$= 0 \text{ when } 3200x = 800$$

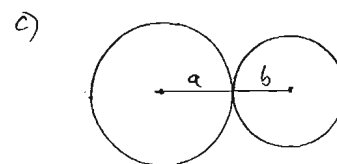
$$\therefore x = \frac{800}{3200}$$

$$= \frac{1}{4} \rightarrow ①$$

$$\frac{d^2E}{dx^2} = -3200$$

$$< 0 \rightarrow ①$$

$$\therefore \text{a maximum in } E \text{ when } x = \frac{1}{4} \text{ min} \rightarrow ①$$



$$a + b = 16$$

$$\therefore b = 16 - a$$

$$A = \pi a^2 + \pi b^2$$

$$= \pi a^2 + \pi (16 - a)^2 \rightarrow ①$$

$$= \pi [a^2 + 256 - 32a + a^2]$$

$$= \pi [2a^2 - 32a + 256]$$

$$= 2\pi [a^2 - 16a + 128]$$

$$\frac{dA}{da} = 2\pi (2a - 16)$$

$$\frac{dA}{da} = 4\pi (a - 8) \rightarrow ①$$

$$= 0 \text{ when } a = 8$$

$$\therefore b = 8$$

$$\frac{d^2A}{da^2} = 4\pi$$

$$\frac{d^2A}{da^2} > 0 \rightarrow ①$$

$$\therefore \text{a minimum area when } a = 8$$

$$\text{and } b = 8 \text{ cm.}$$

$$\therefore \text{Minimum area} = 2 \times 64\pi \rightarrow ①$$

$$= 128\pi \text{ cm}^2$$

Question 3

$$a) (i) \int (2x+3)^2 dx = \frac{(2x+3)^3}{3 \times 2} + C \rightarrow ①$$

$$= \frac{1}{6} (2x+3)^3 + C$$

$$(ii) \int \frac{1}{x^2} dx = \int x^{-2} dx \rightarrow ①$$

$$= \frac{x^{-1}}{-1} + C \rightarrow ①$$

$$= -\frac{1}{x} + C$$

$$b) (i) A \approx \frac{h}{2} [(y_0 + y_n) + 2(y_1 + y_2 + \dots + y_{n-1})] \rightarrow ①$$

$$\approx \frac{0.4}{2} [(2+0) + 2(1.96 + 1.83 + 1.60 + 1.20)] \rightarrow ①$$

$$\approx 0.2 [2 + 2 \times 6.59]$$

$$\approx 3.036$$

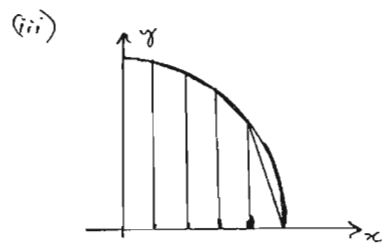
$$\approx 3.04 \text{ units}^2 \rightarrow ①$$

$$(ii) A = \frac{\pi r^2}{4}$$

$$= \frac{\pi (4)^2}{4}$$

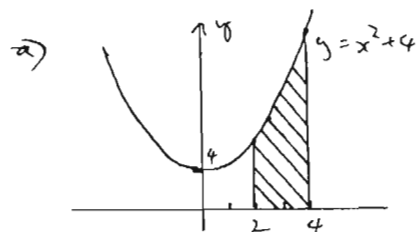
$$\rightarrow ① \leftarrow = \pi \text{ units}^2$$

$$= 3.14159265 \dots \text{ units}^2$$



① ← Approximation is an under-estimate because the trapezia are all within the quadrant.
 $\therefore$ the sum of the trapezia is less than the area of the quadrant.

Question 4



$$A = \int_2^4 x^2 + 4 dx$$

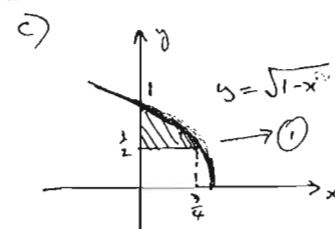
$$= \left[\frac{x^3}{3} + 4x \right]_2^4 \rightarrow ①$$

$$= \left(\frac{64}{3} + 16 \right) - \left(\frac{8}{3} + 8 \right)$$

$$= \frac{56}{3} + 8$$

$$= \frac{80}{3} \text{ units}^2$$

$$= 26 \frac{2}{3} \text{ units}^2 \rightarrow ①$$



$$A = \int_{1/2}^1 1 - y^2 dy \rightarrow ①$$

$$= \left[y - \frac{y^3}{3} \right]_{1/2}^1$$

$$= \left[\left(1 - \frac{1}{3} \right) - \left(\frac{1}{2} - \frac{1}{24} \right) \right]$$

$$= 1 - \frac{1}{3} - \frac{1}{2} + \frac{1}{24}$$

$$= \frac{5}{24} \rightarrow ①$$

$$y = \sqrt{1-x}$$

$$\therefore y^2 = 1-x$$

$$\therefore x = 1-y^2$$

$$\text{When } x = \frac{3}{4},$$

$$\frac{3}{4} = 1-y^2$$

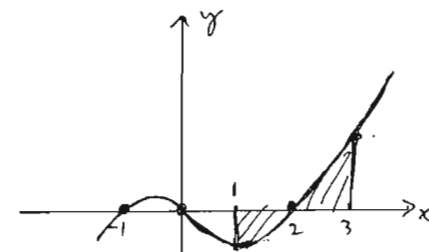
$$\therefore y^2 = 1 - \frac{3}{4}$$

$$= \frac{1}{4}$$

$$\therefore y = \frac{1}{2}$$

$$b) y = x(x+1)(x-2)$$

$$= 0 \text{ when } x = 0, -1, 2 \rightarrow ①$$



$$\text{When } x = 1,$$

$$y = 1 \times 2 \times -1$$

$$< 0$$

$$\text{When } x = 3,$$

$$y = 3 \times 4 \times 1$$

$$> 0$$

$$\therefore \text{Area} = \left| \int_1^2 x(x+1)(x-2) dx \right|$$

$$+ \int_2^3 x(x+1)(x-2) dx \rightarrow ①$$

$$= \left| \int_1^2 (x^3 - x^2 - 2x) dx \right|$$

$$+ \int_2^3 (x^3 - x^2 - 2x) dx$$

$$= \left| \left[\frac{x^4}{4} - \frac{x^3}{3} - x^2 \right]_1^2 \right| \rightarrow ①$$

$$+ \left[\frac{x^4}{4} - \frac{x^3}{3} - x^2 \right]_2^3$$

$$= \left| \left[4 - \frac{8}{3} - 4 \right] - \left[\frac{1}{4} - \frac{1}{3} - 1 \right] \right|$$

$$+ \left[\frac{81}{4} - 9 - 9 \right] - \left[4 - \frac{8}{3} - 4 \right]$$

$$= \left| -\frac{8}{3} - \frac{1}{4} + \frac{1}{3} + 1 \right| + \frac{9}{4} + \frac{8}{3}$$

$$= \left| -\frac{19}{12} \right| + \frac{59}{12} \rightarrow ①$$

$$= 78 = 6.5 \dots \text{ units}^2$$

Question 5

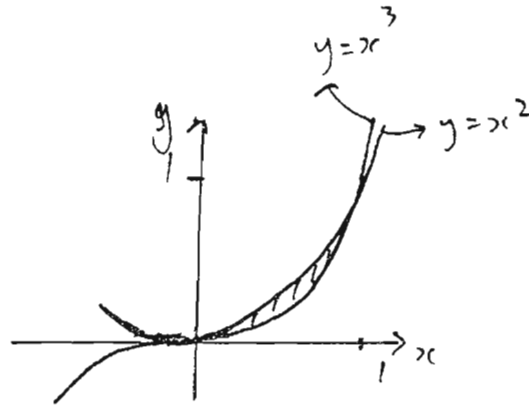
a) $y = x^3$ and $y = x^2$

let $x^3 = x^2$

$\therefore x^3 - x^2 = 0$

$\therefore x^2(x-1) = 0$

$\therefore x = 0, 1 \rightarrow \textcircled{1}$



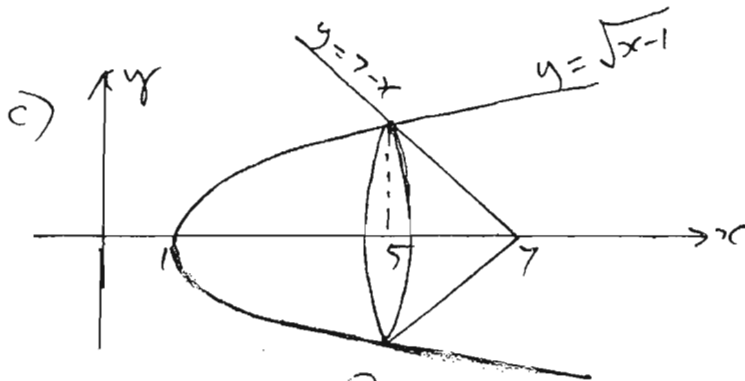
$A = \int_0^1 x^2 dx - \int_0^1 x^3 dx \rightarrow \textcircled{1}$

$= \int_0^1 (x^2 - x^3) dx$

$= \left[\frac{x^3}{3} - \frac{x^4}{4} \right]_0^1$

$= \left(\frac{1}{3} - \frac{1}{4} \right) - (0)$

$= \frac{1}{12} \text{ unit}^2 \rightarrow \textcircled{1}$



$V = \pi \int_1^5 (\sqrt{x-1})^2 dx + \pi \int_5^7 (7-x)^2 dx \rightarrow \textcircled{1}$

$= \pi \int_1^5 (x-1) dx + \pi \int_5^7 (49 - 14x + x^2) dx$

$= \pi \left[\frac{x^2}{2} - x \right]_1^5 + \pi \left[49x - 7x^2 + \frac{x^3}{3} \right]_5^7 \rightarrow \textcircled{1}$

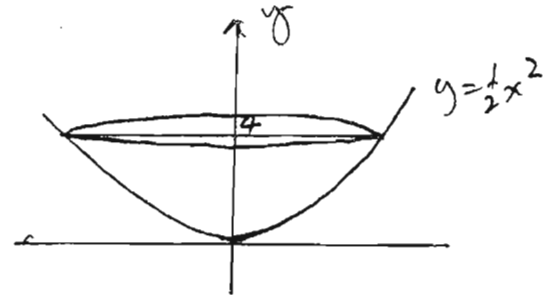
$= \pi \left[\left(\frac{25}{2} - 5 \right) - \left(\frac{1}{2} - 1 \right) \right] + \pi \left[\left(343 - 343 + \frac{343}{3} \right) - \left(245 - 175 + \frac{125}{3} \right) \right]$

$= \pi \left[8 \right] + \pi \left[\frac{343}{3} - \frac{343}{3} \right]$

$= \frac{32\pi}{3} \text{ units}^3 \rightarrow \textcircled{1}$

$\approx 33.51 \text{ u}^3$

b)



$y = \frac{1}{2}x^2$

$\therefore x^2 = 2y$

$\therefore x = \sqrt{2y} \rightarrow \textcircled{1}$

$\therefore V = \pi \int_0^4 (\sqrt{2y})^2 dy \rightarrow \textcircled{1}$

$= \pi \int_0^4 2y dy$

$= \pi \left[y^2 \right]_0^4$

$= \pi (16 - 0) \rightarrow \textcircled{1}$

$= 16\pi \text{ units}^3$